

# Data Science/Data Analytics

## Prerequisites:

Basic Class for Computer Science

Introduction to Data Science (Application,tools,scope,life cycle,etc)

## Topics Covered:

- **Python Programming**
  - Basics
  - Loops and control flows
  - OOPS
  - Modules/Libraries
  - File Handling
- **R**
  - Basic R
  - Data Manipulation
  - Data Visualization
- **Mathematics and Statistics**
  - Linear Algebra (Vectors, Matrices, Eigenvalues)
  - Probability Theory
  - Descriptive Statistics (Mean, Median, Mode, Variance, Standard Deviation)
  - Inferential Statistics (Hypothesis Testing, p-values, Confidence Intervals)
  - Mathematical Modelling
  - Statistical Analysis
- **Data Preprocessing**
  - Data Cleaning (Handling Missing Data, Duplicates)
  - Data Transformation (Scaling, Encoding)
  - Feature Engineering
  - Handling Outliers
- **Data Visualization**
  - Matplotlib ,Seaborn (Python)
  - ggplot2 (R)
  - Plotly, Tableau, Power BI
  - Dashboards & Storytelling with Data
- **Exploratory Data Analysis (EDA)**
  - Understanding Data Distributions
  - Correlation & Causation
  - Identifying Trends and Patterns

- **Database**
  - SQL Basics (Queries, Joins, Aggregations)
  - Advanced SQL (CTEs, Window Functions)
  - NoSQL Databases (MongoDB)
  - Data Warehousing
- **Machine Learning**
  - **Supervised Learning**
    - Linear Regression
    - Logistic Regression
    - Decision Trees & Random Forests
    - Support Vector Machines (SVM),etc
  - **Unsupervised Learning**
    - Clustering (K-Means, Hierarchical)
    - Principal Component Analysis (PCA)
    - Anomaly Detection
  - **Model Evaluation & Tuning**
    - Cross-Validation
    - Hyperparameter Tuning (Grid Search, Random Search)
    - Bias-Variance Tradeoff
- **Deep learning**
  - Introduction to Neural Networks
  - Artificial Neural Networks (ANNs)
  - Convolutional Neural Networks (CNNs)
  - Recurrent Neural Networks (RNNs)
  - Transformers (BERT, GPT),etc
- **Natural language processing**
  - Text Processing (Tokenization, Stemming, Lemmatization)
  - Sentiment Analysis
  - Named Entity Recognition (NER)
  - Topic Modeling
- **Computer vision**
  - Image Processing
  - Object Detection and identifications
  - Face Recognition
  - Biometrics
  - OCR,etc
- **BigData and Cloud Computing**
  - Hadoop & Spark

- Cloud Platforms (AWS, Azure, Google Cloud)
- Data Pipelines & Deployment
- **TimeSeries Analysis**
  - ARIMA & SARIMA Models
  - Facebook Prophet
  - Forecasting Techniques
- **Model Deployment**
  - Deploying Models using Flask/Django
  - Docker & Kubernetes
  - CI/CD for Machine Learning
  - MLOps

### **Technologies and Tools:**

- Programming Language (Python,R,SQL,Scala,Julia)
- Data Manipulation & Analysis (Pandas ,Numpy,Dask,Polars,OpenRefine)
- Data Visualization (Matplot,Seaborn,Plotly,PowerBI,Tableau)
- AI & ML(Scikit Learn,TensorFlow,H2O.ai,XGBoost,PyTorch)
- BigData(Apache Hadoop,Spark,,Dask,Google BigQuery)
- Database(MySql,PostgreSQL,MongoDB,GoogleFirebase,Amazon S3)
- Cloud (Google VertexAI,AWS SageMaker,Azure,MLflow,KuberFlow)
- DataEngineering (Apache Airflow, Talend,Apache Kafka, Data Build Tool(DBT))
- NLP(spacy,HuggingFace,NLTK,TextBlob)
- Git
- AutoML& No-Code ML